

Advanced Machine Learning



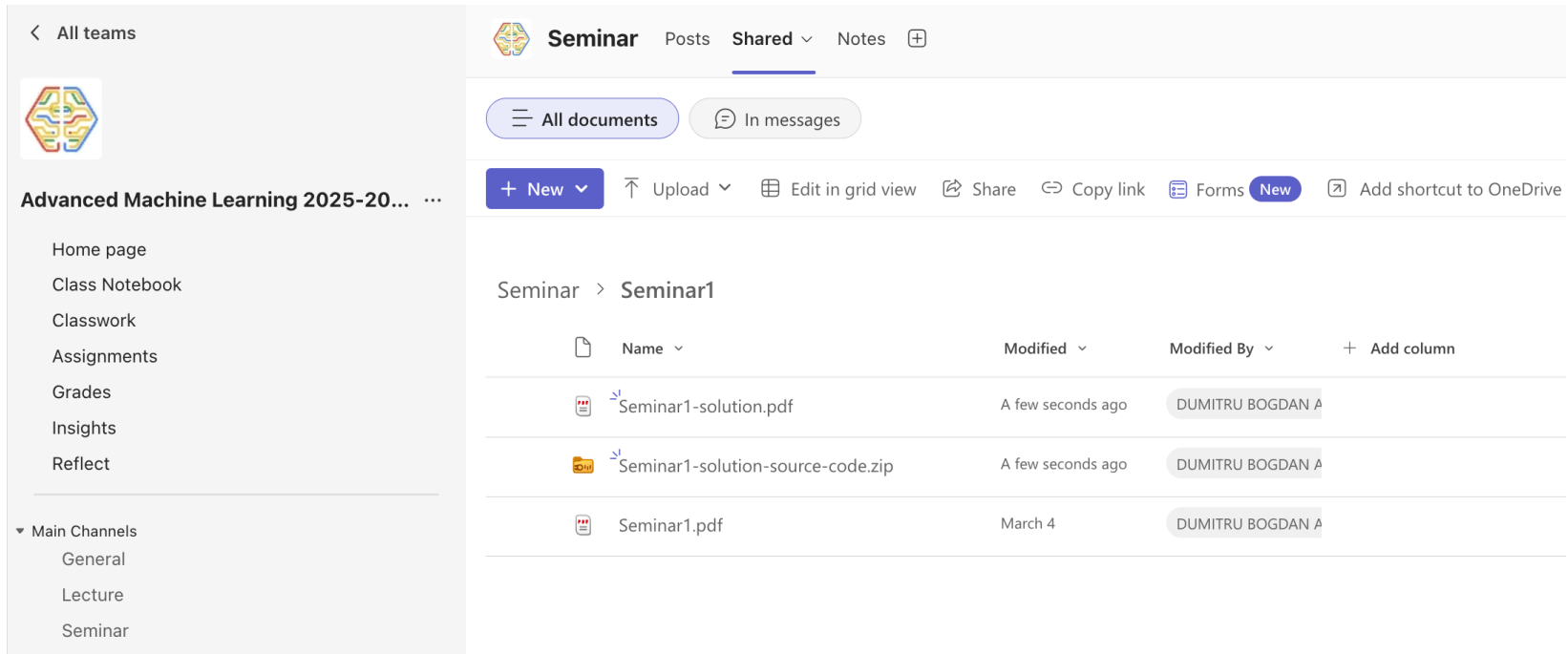
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Administrative

- I've uploaded solution/source files for Seminar 1



The screenshot shows a Microsoft Teams interface for a channel named 'Seminar'. On the left is a sidebar with navigation options: 'All teams', a team icon, 'Advanced Machine Learning 2025-20...', 'Home page', 'Class Notebook', 'Classwork', 'Assignments', 'Grades', 'Insights', 'Reflect', 'Main Channels' (with sub-items 'General', 'Lecture', 'Seminar'), and 'More options'. The main area shows the 'Seminar' channel with tabs for 'Posts', 'Shared' (selected), and 'Notes'. Below the tabs are buttons for 'All documents' and 'In messages'. A toolbar contains '+ New', 'Upload', 'Edit in grid view', 'Share', 'Copy link', 'Forms', 'New' (in a blue bubble), and 'Add shortcut to OneDrive'. Below the toolbar, the breadcrumb 'Seminar > Seminar1' is shown. A table lists shared files:

Name	Modified	Modified By
Seminar1-solution.pdf	A few seconds ago	DUMITRU BOGDAN A
Seminar1-solution-source-code.zip	A few seconds ago	DUMITRU BOGDAN A
Seminar1.pdf	March 4	DUMITRU BOGDAN A

- seminar 2 class this week, 5 exercises
- seminar 2 also next week (Monday + Thursday)

Administrative

- I've uploaded a file with attendance for Seminar 1

< All teams



Advanced Machine Learning 2025-20... ..

Home page

Class Notebook

Classwork

Assignments

Grades

Insights

Reflect

▼ Main Channels

General

Lecture

Seminar

 **Seminar** Posts **Shared** ▾ Notes 

 All documents

 In messages

 New ▾

 Upload ▾

 Edit in grid view

 Share

 Copy link

 Forms **New**

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Seminar

 Name ▾	Modified ▾	Modified By ▾	+ Add column
  AML-2025-2026-BONUS-SEMINAR.pdf	A few seconds ago	DUMITRU BOGDAN A	
 Seminar1	March 4	DUMITRU BOGDAN A	
  Seminar2	6 hours ago	DUMITRU BOGDAN A	

Today's lecture: Overview

- The general PAC learning definition (agnostic PAC)
- Uniform convergence
- The No-Free-Lunch theorem

PAC vs. Agnostic PAC learning

	PAC	Agnostic PAC
Distribution	$\mathcal{D}$ over $\mathcal{X}$	$\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$
Truth	$f \in \mathcal{H}$	not in class or doesn't exist
Risk	$L_{\mathcal{D},f}(h) = \mathcal{D}(\{x : h(x) \neq f(x)\})$	$L_{\mathcal{D}}(h) = \mathcal{D}(\{(x, y) : h(x) \neq y\})$
Training set	$(x_1, \dots, x_m) \sim \mathcal{D}^m$ $\forall i, y_i = f(x_i)$	$((x_1, y_1), \dots, (x_m, y_m)) \sim \mathcal{D}^m$
Goal	$L_{\mathcal{D},f}(A(S)) \leq \epsilon$	$L_{\mathcal{D}}(A(S)) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon$

The general PAC learning problem

- we wish to Probably Approximately solve:

$$\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) \quad \text{where} \quad L_{\mathcal{D}}(h) \stackrel{\text{def}}{=} \mathbb{E}_{z \sim \mathcal{D}} [\ell(h, z)]$$

- learner knows $\mathcal{H}$, $Z = \mathcal{X} \times \mathcal{Y}$ and loss function ℓ
- learner receives accuracy parameter ϵ and confidence parameter δ
- learner can decide on training set size m based on ϵ, δ
- learner doesn't know $\mathcal{D}$ (over Z) but can sample S from $\mathcal{D}^m$
- using S the learner outputs some hypothesis $A(S) = h_S$
- we want that with probability at least $1 - \delta$ over the choice of S , the following would hold:

$$L_{\mathcal{D}}(A(S)) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon$$

Formal definition

A hypothesis class $\mathcal{H}$ is called *agnostic PAC learnable* if there exists a function $m_{\mathcal{H}}: (0,1)^2 \rightarrow \mathbb{N}$ and a learning algorithm A with the following property:

- for every $\varepsilon > 0$ (*accuracy* $\rightarrow$ “approximately correct”)
- for every $\delta > 0$ (*confidence* $\rightarrow$ “probably”)
- for every distribution $\mathcal{D}$ over $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$

when we run the learning algorithm A on a training set S , consisting of $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ examples sampled i.i.d. from $\mathcal{D}$ the algorithm A returns a hypothesis $A(S)$ from $\mathcal{H}$ such that, with probability at least $1-\delta$ (over the choice of examples) it holds that: $L_{\mathcal{D}}(A(S)) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \varepsilon$

$$P_{S \sim \mathcal{D}^m} (L_{\mathcal{D}}(h_S) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \varepsilon) \geq 1 - \delta \Leftrightarrow P_{S \sim \mathcal{D}^m} (L_{\mathcal{D}}(h_S) > \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \varepsilon) < \delta$$

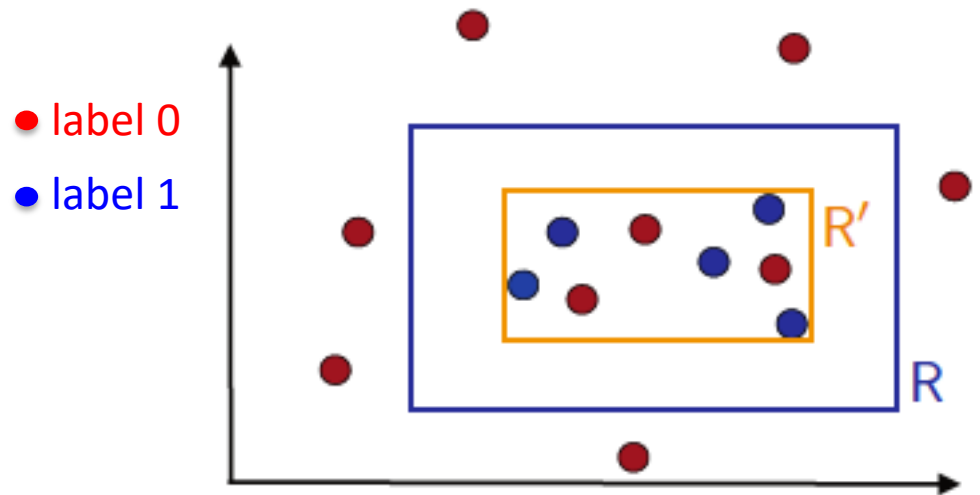
- if the realizability assumption holds, agnostic PAC = PAC
- in agnostic PAC learning, a learner can still declare success if its error is not much larger than the best error achievable by a predictor from the class $\mathcal{H}$.

Learning in the presence of noise - rectangles

- $\mathcal{X} = \mathbb{R}^2$ points in the plane
- $\mathcal{H}$ = set of all axis-aligned rectangle lying in $\mathbb{R}^2$
- each concept $h \in \mathcal{H}$ is an indicator function of a rectangle
- the learning problem consists of determining with small error a target axis-aligned rectangle using the labeled training sample
- the training points received by the learner are subject to noise:
 - points negatively labeled are unaffected by noise
 - the label of a positive training points is randomly flipped to negative with probability $0 < \eta < \frac{1}{2}$ (η is unknown)

$\mathcal{H}$ is agnostic PAC learnable

$$\min_h L_{\mathcal{D}}(h) = \eta \times \mathcal{D}(R)$$



A note of Caution

The fact that $\mathcal{H}$ is agnostically PAC learnable using the ERM paradigm doesn't mean that the result is any good.

It only means that you can be reasonable sure the ERM paradigm gives you a result that is close to the optimal result.

If the optimal result is bad (because, for example, the hypothesis class $\mathcal{H}$ fits the data really badly) the ERM paradigm will also give you a bad result.

PAC doesn't tell you that your hypothesis class $\mathcal{H}$ fits the data well, it only tells you that, if it fits well, the ERM paradigm will probably give you a reasonable good hypothesis.

Beyond the general PAC learning definition

- the definition of the general PAC learning tells us:
 - when we consider we can learn something
- the definition of the general PAC learning doesn't tell us:
 - what we can learn
 - how we learn
- discover what can be general PAC-learned and how

Uniform Convergence

Sufficient learning condition for agnostic PAC learnability

- given $\mathcal{H}$, the $\text{ERM}_{\mathcal{H}}$ learning paradigm works as follows:
 - based on a received training sample S of examples draw i.i.d from an unknown distribution $\mathcal{D}$ over a domain $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, $\text{ERM}_{\mathcal{H}}$ evaluates the risk (error) of each h in $\mathcal{H}$ on S and outputs a member $h_S = \text{ERM}_{\mathcal{H}}(S)$ that minimizes the empirical error $L_S(h_S)$;
 - we want that h_S will generalize wrt true data probability distribution $\mathcal{D}$, i.e $L_{\mathcal{D}}(h_S)$ is small;
 - it suffices to ensure that the empirical risks of all h in $\mathcal{H}$ are good approximations of their true risk
- we need that *uniformly* over all hypothesis h in the hypothesis class $\mathcal{H}$, the empirical risk based on S will be close to true risk for all possible probability distributions $\mathcal{D}$ over the domain $\mathcal{Z}$

ϵ - Representative

- how well you can learn a hypothesis depends on the quality of that sample:
 - you can't learn anything from a bad sample
 - a bad sample will make a bad hypothesis to look good and a good one to look bad
- when is a sample good?
 - a sample is good if the estimated quality (the loss) of a hypothesis on that sample is very close to its true error

Definition (ϵ – representative sample)

A sample S is called ϵ – representative wrt domain $Z = \mathcal{X} \times \mathcal{Y}$, hypothesis class $\mathcal{H}$, loss function l and distribution $\mathcal{D}$ if:

$$\forall h \in \mathcal{H}, \quad |L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon.$$

$$L_{\mathcal{D}}(h) \stackrel{\text{def}}{=} \mathbb{E}_{z \sim \mathcal{D}} [l(h, z)] \quad L_S(h) = \frac{1}{m} \sum_{z \in S} l(h, z)$$

ϵ – Representative Samples are Good

Lemma

Let S be a sample that is $\epsilon/2$ – representative wrt domain $\mathcal{Z}$, hypothesis class $\mathcal{H}$, loss function ℓ and distribution $\mathcal{D}$. Then any output of $A(S) = \text{ERM}_{\mathcal{H}}(S)$ i.e any $h_S \in \text{argmin}_h L_S(h)$ satisfies:

$$L_{\mathcal{D}}(A(S)) \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon$$

Proof

$$L_{\mathcal{D}}(h_S) \leq L_S(h_S) + \epsilon/2 \leq \min_h L_S(h) + \epsilon/2 \leq \min_h L_{\mathcal{D}}(h) + \epsilon/2 + \epsilon/2$$

S is $\epsilon/2$ – representative sample

Uniform convergence

If ϵ -representative samples allows us to learn as good as possible, we can agnostically PAC learn if we can guarantee that we will almost always get (with probability $1 - \delta$) ϵ -representative sample.

Definition (*uniform convergence*)

A hypothesis class $\mathcal{H}$ has the *uniform convergence property* wrt a domain Z , loss function l if:

- there exists a function $m_H^{UC} : (0,1)^2 \rightarrow \mathbb{N}$
- such that for all $(\epsilon, \delta) \in (0,1)^2$
- and for any probability distribution $\mathcal{D}$ over Z

if S is a sample of $m \geq m_H^{UC}(\epsilon, \delta)$ examples drawn i.i.d. according to $\mathcal{D}$, then, with probability of at least $1 - \delta$, S is ϵ -representative.

The term *uniform* refers to having a fixed sample size that works for all members of $\mathcal{H}$ and over all possible probability distributions $\mathcal{D}$ over the domain Z

A tool to prove PAC learnability

- uniform convergence serves as a tool to prove that we can PAC learn a hypothesis class $\mathcal{H}$

Corollary

If hypothesis class $\mathcal{H}$ has the uniform convergence property with function m_H^{UC} then $\mathcal{H}$ is agnostically PAC learnable with the sample complexity:

$$m_H(\varepsilon, \delta) \leq m_H^{UC}(\varepsilon / 2, \delta)$$

Moreover, the $\text{ERM}_{\mathcal{H}}$ paradigm is a successful agnostic PAC learner for $\mathcal{H}$.

Finite classes are agnostic PAC learnable

Theorem

Let $\mathcal{H}$ be a finite hypothesis class, let $\mathcal{Z}$ be a domain and let $\ell: \mathcal{H} \times \mathcal{Z} \rightarrow [0,1]$ be a loss function. Then $\mathcal{H}$ has the uniform convergence property with sample complexity:

$$m_{\mathcal{H}}^{UC}(\epsilon, \delta) \leq \left\lceil \frac{\log(2|\mathcal{H}|/\delta)}{2\epsilon^2} \right\rceil$$

Moreover, the class $\mathcal{H}$ is agnostically PAC learnable using the ERM paradigm with sample complexity:

$$m_{\mathcal{H}}(\epsilon, \delta) \leq m_{\mathcal{H}}^{UC}(\epsilon/2, \delta) \leq \left\lceil \frac{2\log(2|\mathcal{H}|/\delta)}{\epsilon^2} \right\rceil$$

Proof - Finite classes are agnostic PAC learnable

- uniform convergence serves as a tool to prove that we can PAC learn a hypothesis class $\mathcal{H}$
- to prove that finite hypothesis classes have the uniform convergence property, we need to:
 - for fixed ϵ and δ
 - find a sample size m
 - such that for any distribution $\mathcal{D}$ over $\mathcal{Z}$
 - and a sample $S = (z_1, z_2, \dots, z_m)$ of examples i.i.d from $\mathcal{D}$
 - with probability at least $1 - \delta$
 - it holds that for all $h \in \mathcal{H}$ $|L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon$.

That is: $\mathcal{D}^m(\{S : \forall h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| \leq \epsilon\}) \geq 1 - \delta$.



$$\mathcal{D}^m(\{S : \exists h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) < \delta.$$

Proof - union bound

$$\{S : \exists h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\} = \cup_{h \in \mathcal{H}} \{S : |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\},$$

Use the union bound to obtain:

$$\mathcal{D}^m(\{S : \exists h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) \leq \sum_{h \in \mathcal{H}} \mathcal{D}^m(\{S : |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}).$$

For a sufficiently large m , each summand of the right-hand side of this inequality is small enough.

Show that for any fixed hypothesis h (which is chosen in advance prior to the sampling of the training set), the gap between the true and empirical risks, $|L_S(h) - L_{\mathcal{D}}(h)|$, is likely to be small.

Proof - Hoeffding's inequality

Lemma (Hoeffding's Inequality). *Let $\theta_1, \dots, \theta_m$ be a sequence of i.i.d. random variables and assume that for all i , $\mathbb{E}[\theta_i] = \mu$ and $\mathbb{P}[a \leq \theta_i \leq b] = 1$. Then, for any $\epsilon > 0$*

$$\mathbb{P} \left[\left| \frac{1}{m} \sum_{i=1}^m \theta_i - \mu \right| > \epsilon \right] \leq 2 \exp \left(-2m \epsilon^2 / (b - a)^2 \right).$$

Apply in our case by setting:

$$\theta_i = l(h, z_i) \quad L_S(h) = \frac{1}{m} \sum_{z \in S} l(h, z) = \frac{1}{m} \sum_i^m \theta_i \quad L_D(h) = \mu \quad a = 0, b = 1$$

Then, we have:

$$\mathcal{D}^m(\{S : |L_S(h) - L_D(h)| > \epsilon\}) = \mathbb{P} \left[\left| \frac{1}{m} \sum_{i=1}^m \theta_i - \mu \right| > \epsilon \right] \leq 2 \exp \left(-2m \epsilon^2 \right).$$

Proof - final step

$$\begin{aligned}\mathcal{D}^m(\{S : \exists h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) &\leq \sum_{h \in \mathcal{H}} 2 \exp(-2m\epsilon^2) \\ &= 2|\mathcal{H}| \exp(-2m\epsilon^2)\end{aligned}$$

Choose $m \geq \frac{\log(2|\mathcal{H}|/\delta)}{2\epsilon^2}$

Then, we have:

$$\mathcal{D}^m(\{S : \exists h \in \mathcal{H}, |L_S(h) - L_{\mathcal{D}}(h)| > \epsilon\}) \leq \delta.$$

Beyond the result

By going from realizability to agnostic, we go:

- from $m_{\mathcal{H}}(\epsilon, \delta) \leq \left\lceil \frac{\log(|\mathcal{H}|/\delta)}{\epsilon} \right\rceil$ (lecture 2)
- to $m_{\mathcal{H}}(\epsilon, \delta) \leq m_{\mathcal{H}}^{UC}(\epsilon/2, \delta) \leq \left\lceil \frac{2\log(2|\mathcal{H}|/\delta)}{\epsilon^2} \right\rceil$

The denominator goes from ϵ to ϵ^2 , which means that for the same of accuracy the minimal sample size grows by a factor of $1/\epsilon$.

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